

# CSA 0378-DATA STRUCTURES

## 13. QUEUE USING LINKED LIST

```
main.c
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 struct node
5 {
6     int data;
7     struct node *next;
8 };
9
10 struct node *front = NULL;
11 struct node *rear = NULL;
12
13 void enqueue(int value)
14 {
15     struct node *newNode;
16
17     newNode = (struct node *)malloc(sizeof(struct node));
18     newNode->data = value;
19     newNode->next = NULL;
20
21     if(rear == NULL)
22     {
23         front = rear = newNode;
24     }
25     else
26     {
27         rear->next = newNode;
28         rear = newNode;
29     }
30
31     printf("%d inserted into queue\n", value);
32 }
33
34 void dequeue()
35 {
36     struct node *temp;
```

Output

```
1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter Choice: 1
Enter Value: 2
2 inserted into queue

1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter Choice: 3
Queue Elements:
2

1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter Choice: 1
Enter Value: 2
2 inserted into queue

1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter Choice: 3
Queue Elements:
2 2

1. Enqueue
2. Dequeue
3. Display
```

```
main.c
34 void dequeue()
35 {
36     struct node *temp;
37
38     if(front == NULL)
39     {
40         printf("Queue Underflow\n");
41         return;
42     }
43
44     temp = front;
45     printf("Deleted Element: %d\n", front->data);
46
47     front = front->next;
48
49     if(front == NULL)
50         rear = NULL;
51
52     free(temp);
53 }
54
55 void display()
56 {
57     struct node *temp;
58
59     if(front == NULL)
60     {
61         printf("Queue is Empty\n");
62         return;
63     }
64
65     printf("Queue Elements:\n");
66
67     temp = front;
68     while(temp != NULL)
```

Output

```
1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter Choice: 1
Enter Value: 2
2 inserted into queue

1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter Choice: 3
Queue Elements:
2

1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter Choice: 1
Enter Value: 2
2 inserted into queue

1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter Choice: 3
Queue Elements:
2 2

1. Enqueue
2. Dequeue
3. Display
```

```
main.c
78 int main()
79 {
80     int choice, value;
81
82     while(1)
83     {
84         printf("\n1. Enqueue\n");
85         printf("\n2. Dequeue\n");
86         printf("\n3. Display\n");
87         printf("\n4. Exit\n");
88         printf("Enter Choice: ");
89         scanf("%d", &choice);
90
91         if(choice == 1)
92         {
93             printf("Enter Value: ");
94             scanf("%d", &value);
95             enqueue(value);
96         }
97         else if(choice == 2)
98         {
99             dequeue();
100         }
101         else if(choice == 3)
102         {
103             display();
104         }
105         else if(choice == 4)
106         {
107             break;
108         }
109         else
110         {
111             printf("Invalid Choice\n");
112         }
113     }
114 }
```

Output

```
2. Dequeue
3. Display
4. Exit
Enter Choice: 1
Enter Value: 2
2 inserted into queue

1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter Choice: 3
Queue Elements:
2

1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter Choice: 1
Enter Value: 2
2 inserted into queue

1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter Choice: 3
Queue Elements:
2 2

1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter Choice: 1
```